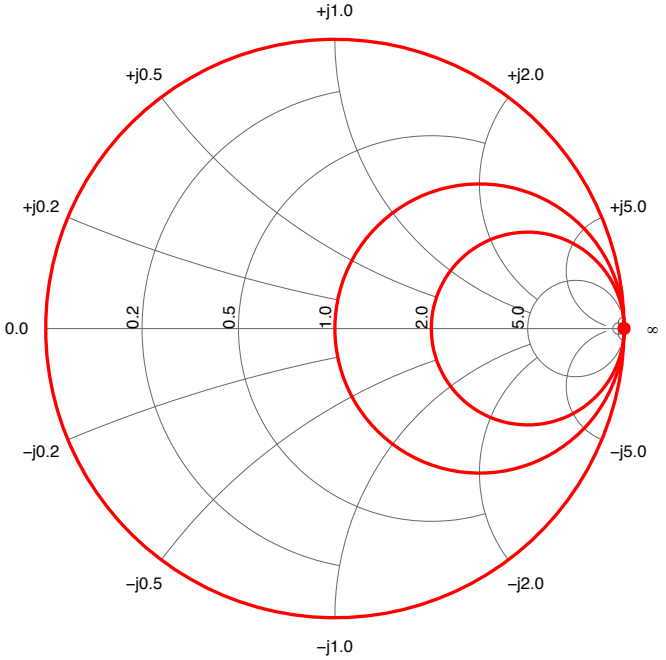
1. (Bonus problem) (10 pts)The red circles in the following figure correspond to normalized impedances $z = \frac{Z}{Z_o} = r + jx$, for fixed values of r. We can fill the following table:

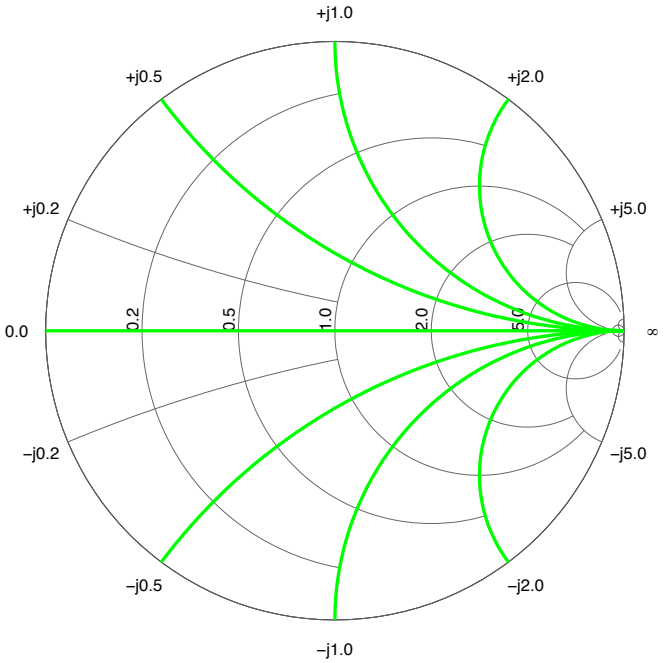
$r =$	center$= \left(\frac{r}{r+1}, 0\right)$	radius$= \frac{1}{r+1}$
0	$(0, 0)$	1
1	$(0.5, 0)$	0.5
2	$(\frac{2}{3}, 0)$	$\frac{1}{3}$
∞	$(1, 0)$	0



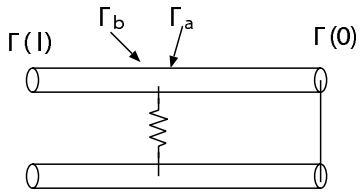
The green circles in the following figure correspond to normalized impedances $z = \frac{Z}{Z_o} = r + jx$, for fixed values of x. We can fill the following table:

$x =$	center$= \left(1, \frac{1}{x}\right)$	radius$= \frac{1}{x}$
0	$(1, \infty)$	∞
± 0.5	$(1, \pm 2)$	± 2
± 1	$(1, \pm 1)$	± 1
± 2	$(1, \pm 0.5)$	± 0.5

Note that the infinite radius for $x = 0$ means that the locus of pure real impedances is the straight line of the real axis.

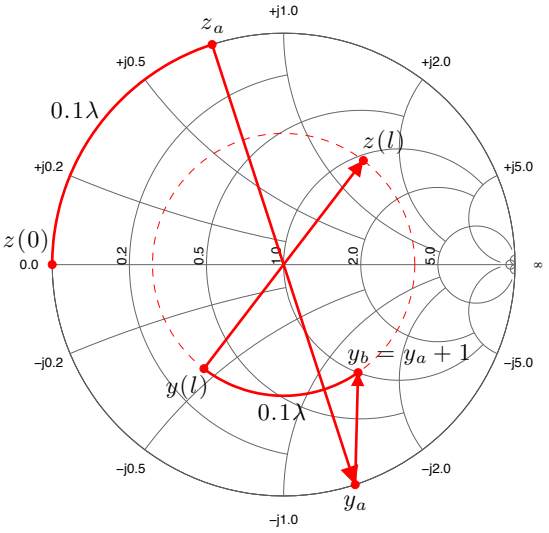


2. The case given in the problem can be display as in the following figure:



Using the Smith chart and working in terms of impedances we find:

- Impedance at the load $Z_L = 0$,
- Normalized input impedance at the load $z_L = \frac{Z_L}{Z_o} = 0$, now rotate from point z_L towards the generator (clockwise) by $d = 0.1\lambda$ until we reach A (at shunt connection)
- Normalized input impedance A at shunt connection is $z_a = j0.727$.
- Normalized input admittance A at shunt connection is $y_a = \frac{1}{z_a} = -j1.376$.
- The normalized shunt resistor is $z_R = R/Z_o = 50/50 = 1\,\Omega$, and thus $y_R = 1\,\Omega$.
- Then, normalized input admittance B at shunt connection is $y_b = y_a + y_R = 1 - j1.376$.
- Normalized input impedance B at shunt connection is given by $z_b = \frac{1}{1+y_a} = 0.345 + j0.476$, now rotate from point z_b towards the generator (clockwise) by $d = 0.1\lambda$ until we reach the generator.
- Normalized impedance at the generator will be $z(l) = 1.074 + j1.425$.
- Impedance at the generator is $Z(l) = 50 \cdot z(l) = 53.71 + j71.23\,\Omega$.
- Normalized admittance at the generator is $y(l) = 1/z(l) = 0.337 + j0.448$.
- Finally, admittance at the generator $Y(l) = \frac{1}{50} \cdot y(l) = (6.75 - j8.95) \times 10^{-3}\,\Omega^{-1}$



3. This problem is very similar to Example 2 of Lecture 37. We can find the distance d_1 by using the Smith chart.

- First, we find the normalized load impedance $z_L = \frac{Z_L}{Z_o}$ on SC,
- Then, we should draw a circle whose radius will be equal to the distance from the center of SC to z_L,

- The point z_L' where the circle intersects the real axis (i.e. where the impedance is purely real),
- Starting at z_L measure the angle you rotate towards the generator (i.e. clockwise) until you reach z_L',
- This angle will give you the distance d_1 in λ-units,
- Finally, the impedance at this point will $Z(d_1) = Z_o \cdot z_L'$.

a) Given that $Z_L = 100\,\Omega$, then $z_L = \frac{Z_L}{Z_o} = 2\,\Omega$. Since z_L is already on real axis, we do not have to find $z_L^{'}$. Thus, we have

$$d_1 = 0\lambda$$

and $Z(d_1) = Z_L = 100\,\Omega$. The characteristic impedance of the quarter-wave transformer is

$$Z_{qo} = \sqrt{Z_o \cdot Z(d_1)} = \sqrt{50 \cdot 100} = 70.71\,\Omega.$$

b) Given that $Z_L = 100 + j100\,\Omega$, then $z_L = \frac{Z_L}{Z_o} = 2 + j2\,\Omega$. Now, we will draw a circle passing through z_L and this circle will intersect the real axis at $z_L^{'} = 4.26$. The angle rotated from z_L to $z_L^{'}$ gives us the distance as

$$d_1 = 0.0413\lambda$$

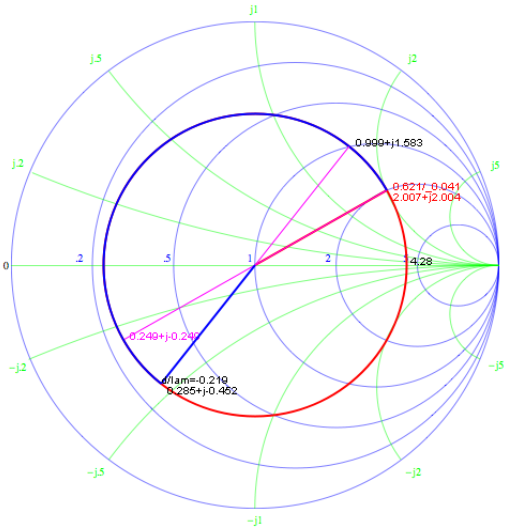
and $Z(d_1) = Z_o \cdot z_L^{'} = 50 \cdot 4.26 = 213\,\Omega$. The characteristic impedance of the quarter-wave transformer is

$$Z_{qo} = \sqrt{Z_o \cdot Z(d_1)} = \sqrt{50 \cdot 213} = 103.19\,\Omega.$$

4. Refering to Example 3 in course notes Lecture 37, d_1 needs to be selected such that $y(d_1) = 1 + jb$, and then the shorted stub which has a purely imaginary impedance will provide $y_s = -jb$, so that $y(d_1) + y_s = 1 + j0$, which means that the impedance is matched at d_1.
Then, the normalized load impedance and admittance are

$$\begin{aligned} z_L &= Z_L/Z_o = 2 + j2 \\ y_L &= 1/z_L = 0.25 - j0.25. \end{aligned}$$

We rotate from point y_L towards the generator (clockwise) until we reach the circle $g = 1$, and the intersection point reads $y(d_1) = 1 + j1.58$.
The rotated distance is $d_1 = 0.219\lambda$, and the admittance of the stub is $y_s = 1 + j0 - y(d_1) = -j1.58$.



To determine the length of the stub, we locate the short point $y = \infty$ and rotate towards the generator (clockwise) until we reach the point $y = -j1.58$. The rotation distance is the length of the stub: $d_s = 0.09\lambda$.

